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Whitepaper

Open Platform Mandate

Why Data and AI Success Depends on Openness and Portability



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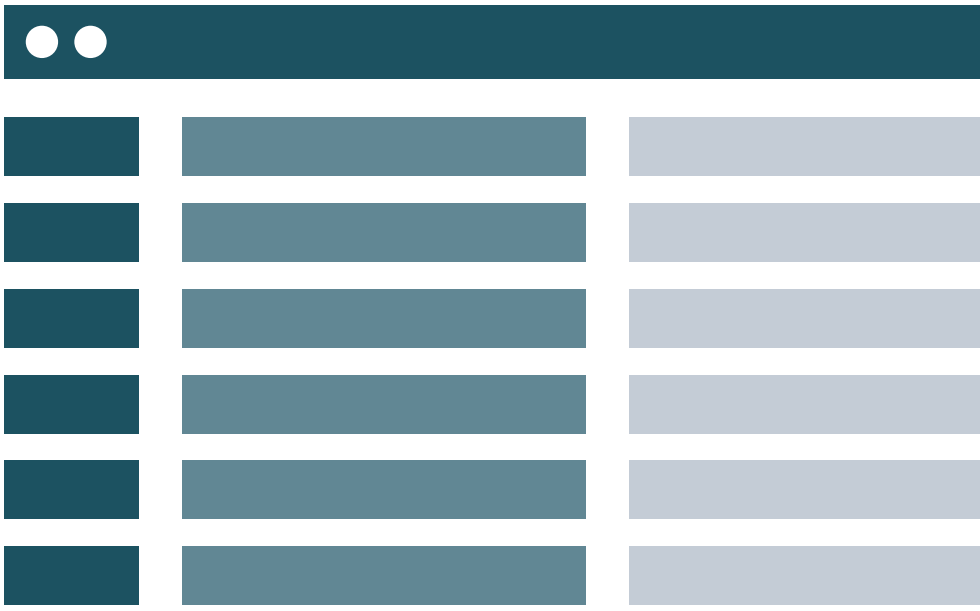
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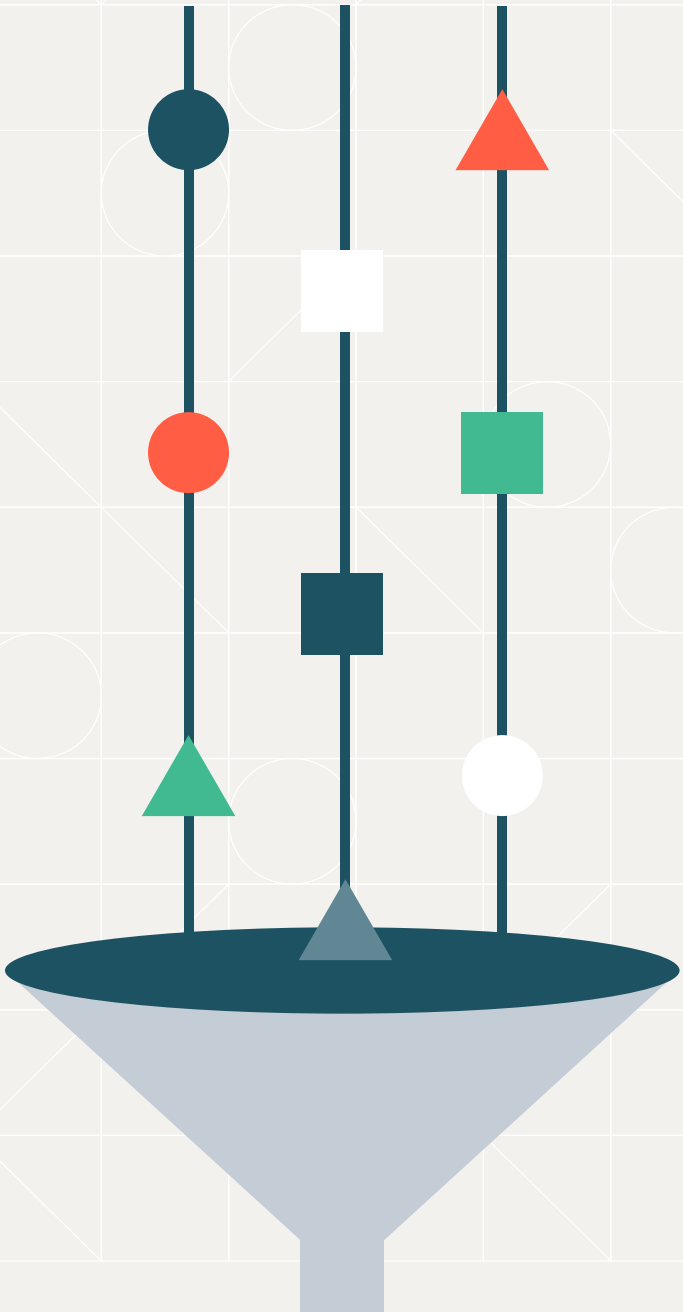
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Executive Summary

As the original creators of Apache Spark™ and leaders in open source innovation, Databricks empowers organizations to build on portable, open foundations while accessing enterprise-grade capabilities. This whitepaper breaks down how embracing portability eliminates vendor lock-in and shows why millions of developers trust solutions built on portable, open standards.



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The exponential growth in data volume and AI capabilities has created unprecedented opportunities for organizations to transform their operations and create value. However, this opportunity comes with significant challenges:

- How can organizations build data and AI infrastructure that remains flexible and future-proof, enabling seamless integration with emerging technologies, supporting diverse workloads and scaling across clouds?
- How can they avoid vendor lock-in while still benefiting from enterprise-grade capabilities such as data governance and security, data portability, and streamlined integration with existing tools and workflows?
- How can they ensure their data and AI infrastructure investments will continue to provide value as technology evolves?

The answer lies in the portability of data, code and workflows — key enablers of operational flexibility and scalability in data and AI initiatives. Leading organizations have discovered that open platforms designed for interoperability and data portability deliver production capabilities while preserving the flexibility and control necessary for long-term success. This approach is crucial as companies scale their data and AI initiatives, where the freedom and ability to adapt to shifting business demands and new technologies impact revenue growth opportunities.

For technical leaders evaluating their options, understanding the practical implications of platform openness — from data portability to ecosystem compatibility — can influence their long-term success. This paper explores how organizations can leverage openness to construct scalable, flexible data and AI infrastructure that evolves with their needs.

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Understanding the True Meaning of "Openness"

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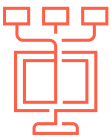
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When evaluating open platforms, technical leaders will explore:

- **Open source:** Core technologies available under permissive licenses that allow modification and redistribution
- **Open standards:** Adherence to widely adopted industry standards rather than proprietary formats
- **Open formats:** Data stored in formats that can be accessed using standard tools and libraries
- **Open ecosystem:** Ability to integrate with and leverage a broad range of tools and technologies

In the context of open data and AI platforms, "portable" encompasses several critical dimensions:



Platforms

Move data and applications between different cloud providers or systems without disruption or loss of functionality



Data

Retain full control over information, which enables organizations to transfer data between platforms or vendors without being locked into proprietary formats or systems



Workflows

Maintain the flexibility to execute processes on any platform to adapt to changing business needs without requiring a complete redesign



Talent

Employees with sought-after technical skills can transition between different organizations or platforms, ensuring teams remain adaptable and resourceful

These four dimensions of portability create a foundation for sustainable, scalable data and AI infrastructure. Organizations that evaluate platforms against these criteria can better assess their long-term viability and alignment with business objectives.

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The Strategic Importance of Portability in Data and AI

The requirement for data portability changes how enterprises build and maintain their data and AI infrastructure. Organizations prioritizing data portability and openness in standards, formats and platforms gain three distinct competitive advantages:

Risk mitigation through independence

Data portability eliminates dependency on single-vendor roadmaps and pricing structures. This independence matters because it gives organizations control over their technology destiny. Moreover, openness allows for community-validated security, transparent development processes and shared governance to ensure reliable, secure platforms that adapt to changing business needs.

Innovation at market speed

Portability accelerates innovation by allowing organizations to gain direct access to the latest advancements across a wide range of tools and technologies. Companies can customize and extend platform capabilities to fit their unique needs without relying on vendor dependencies, fostering greater adaptability and control. Additionally, they benefit from rapid problem-solving through community knowledge and collaboration, drawing on a collective wealth of expertise. This approach also enables seamless integration with emerging tools and technologies — providing the flexibility needed to evolve with industry trends.

Talent advantage in a competitive market

Portability offers distinct advantages for building and scaling technical teams by providing access to a global pool of professionals experienced with multiple tools, platforms and technologies, which broadens talent options and accelerates hiring. With the widespread adoption of portable technologies and open standards, training costs are reduced, allowing teams to onboard more efficiently. Team satisfaction also increases as they work with preferred formats and tools, fostering a productive and motivated environment. Additionally, contributing to and learning from the broader open community encourages continuous growth, knowledge sharing and skill development.

Each of these advantages compounds over time as organizations build deeper expertise with their data platform technologies and participate more actively in the broader technical community.

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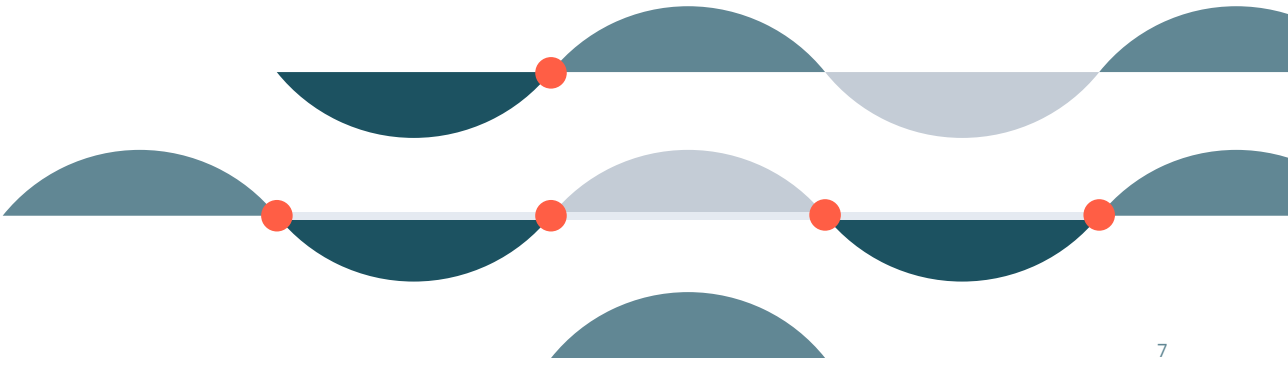
Since our founding by the original creators of Apache Spark, Databricks has maintained a deep commitment to open source development. We've contributed millions of lines of code to the open source community and launched multiple projects that have become industry standards for data and AI.

Built on open source innovation

The Databricks Data Intelligence Platform is fundamentally built on open source technologies, with Apache Spark at its core, reflecting our commitment to openness and transparency. This foundation enables organizations to leverage their existing open source expertise, maintain control over their technology stack and avoid the limitations of proprietary lock-in.

Clear separation of concerns

We maintain transparency boundaries between our open source core and proprietary enhancements: foundational technologies like Apache Spark, Delta Lake and MLflow form the open source core, while enterprise features such as Photon and Delta Sharing add advanced capabilities without compromising portability. Additionally, managed infrastructure and optimization are provided through our cloud services.



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Platform Compatibility and Portability

True platform portability requires more than just using open source technologies — it demands seamless integration with the broader ecosystem of tools and frameworks that data teams rely on daily. Databricks delivers this integration at every layer of the data and AI stack.

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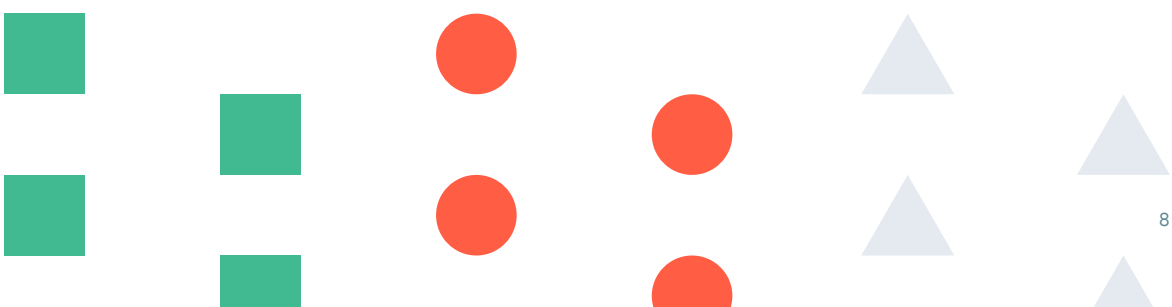
Integration with an open ecosystem

The platform is designed to integrate with popular open source tools and frameworks, empowering organizations to build robust, flexible data and AI solutions by combining industry-standard technologies with Databricks capabilities.

- **Data processing:** Apache Spark, Apache Arrow
- **Machine learning:** TensorFlow, PyTorch, scikit-learn
- **Data formats:** Parquet, Delta Lake, Apache Iceberg™
- **Development tools:** Git, Jupyter, VS Code

Ensuring data portability

Portability isn't just about platform flexibility; it's about maintaining control over data. Databricks supports this goal through a comprehensive commitment to data portability by using standard open formats for data storage and exchange, a cloud-agnostic architecture that avoids vendor lock-in, and clear, documented export procedures. This flexibility is further reinforced by offering multiple deployment models across cloud environments, empowering organizations to move, manage and leverage their data freely for long-term independence and adaptability.



Transitioning Between Platforms

While Databricks strives to be your long-term partner, we believe in giving customers complete freedom of choice. Understanding the practical considerations of platform transitions helps organizations make informed decisions about their data infrastructure.

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Migration considerations

The path to platform transition requires careful evaluation across three critical areas that determine success. Each area presents distinct challenges and opportunities for organizations planning their migration strategy.

DATA FORMATS AND STORAGE

The foundation of any successful migration starts with data. Organizations must consider native support for open formats to prevent future lock-in, assess data export capabilities and their potential impact on operations, and evaluate storage costs and requirements across platforms.

Real-world migrations show that organizations that validate these elements early avoid costly surprises later. For example, teams using open formats like Delta Lake or Apache Iceberg maintain more flexibility in their transition options.

PROCESSING WORKFLOWS

Understanding how existing workflows will transition to a new environment is crucial for shaping both the timeline and the risk associated with migration. Organizations must evaluate the code portability between platforms, consider the performance implications for critical workloads, and assess the resource requirements and optimization needs to ensure a smooth and efficient transition.

Organizations running Apache Spark workloads often find that their code runs with minimal modification on Databricks, while transitions to proprietary platforms may require substantial rewrites.

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OPERATIONAL IMPACT

Successful transitions extend beyond technical considerations and necessitate careful attention to team and business factors. Organizations must assess team expertise with the target technologies, anticipate changes in maintenance overhead and evaluate the total cost of ownership, including any hidden costs.

Leading organizations carefully evaluate these operational factors early in their planning process, as they often have the greatest impact on long-term migration success.

Risk mitigation strategies

Platform transitions carry inherent risks that require careful planning and management. Organizations can protect their operations and ensure successful migrations by focusing on key risk areas:

Data migration — protect your most valuable asset during transition:

- Export data in open, standard formats
- Validate data completeness and integrity
- Maintain clear lineage and governance controls
- Ensure historical data remains accessible

Workload continuity — minimize disruption to business operations:

- Identify platform-specific code and dependencies
- Test performance on new infrastructure
- Plan for service-level agreement changes
- Run parallel environments during the transition

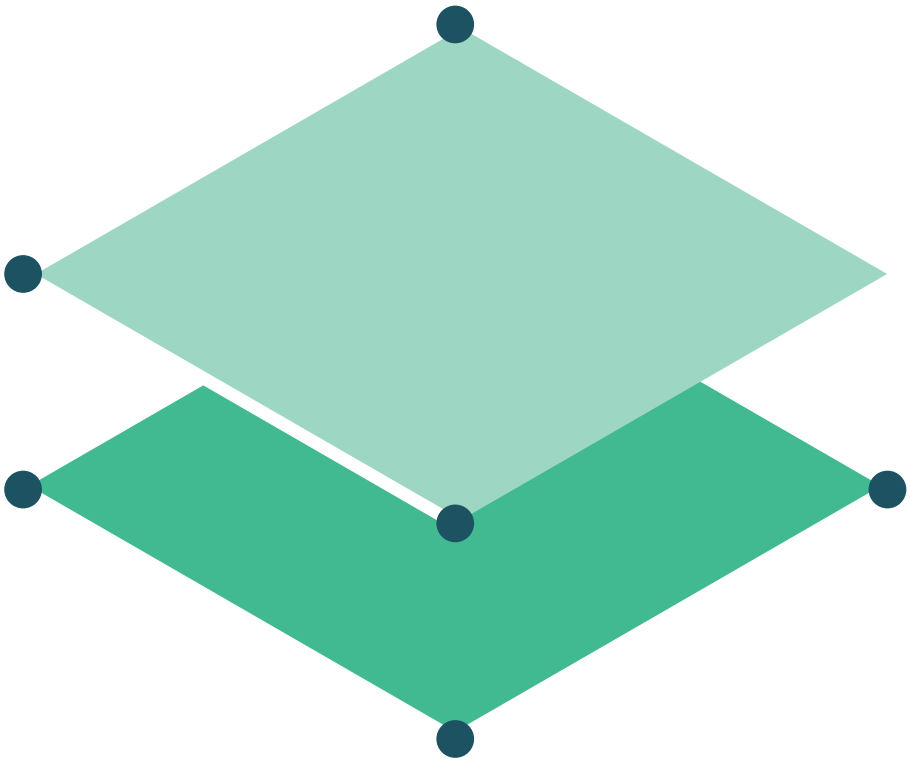
Operational readiness — prepare your team and processes:

- Assess training needs for new technologies
- Update operational procedures and documentation
- Modify monitoring and alerting systems
- Revise backup and disaster recovery plans

Building the Future on Open Foundations

The future of enterprise data and AI belongs to open platforms. As organizations scale their data initiatives and accelerate AI adoption, the ability to maintain flexibility while accessing enterprise capabilities becomes crucial. Portability ensures organizations have the freedom to adapt to changing technologies without being locked into any one solution.

Databricks' commitment to openness reflects our belief in the power of maintaining control of technology and data infrastructure, enabling organizations to innovate faster and scale with ease. For technical leaders charting their data and AI strategy, the path forward is clear. Portable platforms reduce risk, accelerate innovation and attract top talent. They provide the foundation for sustainable growth while ensuring organizations maintain control of their technology destiny.



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Begin your journey with Databricks and portable data and AI:

Visit databricks.com/product/open-source to explore our open source projects and engage with the broader community.

Learn about the Built on Databricks Partner Program that provides access to technical and go-to-market resources to accelerate your development and grow your business.

Explore our Databricks Professional Services offerings and contact us to get started databricks.com/contact.

Databricks is the data and AI company. More than 10,000 organizations worldwide — including Block, Comcast, Condé Nast, Rivian, Shell and over 60% of the Fortune 500 — rely on the Databricks Data Intelligence Platform to take control of their data and put it to work with AI. Databricks is headquartered in San Francisco, with offices around the globe, and was founded by the original creators of Lakehouse, Apache Spark™, Delta Lake and MLflow. To learn more, follow Databricks on LinkedIn, X and Facebook.

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